The first-order heat tent square function has a sharp threshold at $\sigma = 1$

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13 August 2026

Status. Candidate full counterexample; likely valid subject to human review. It answers negatively the all-σ tent/Sobolev equivalence asked by Portal and Veraar [1].

1 The question

For a rough time-dependent divergence-form operator, let $\Gamma(t,s)$ be its evolution family. The source defines

$$\|g\|_{T^{p,2}_\sigma}^p = \int_{\mathbb{R}^d} \left(\int_0^\infty \frac{1}{|B(x,\sqrt{t})|} \int_{B(x,\sqrt{t})} |g(y,t)|^2 \, dy \, \frac{dt}{t^{1+\sigma}} \right)^{p/2} dx$$

and asks, for $\sigma \geq 0$, whether

$$\|(-\Delta)^{\sigma/2} f\|_{L^p(\mathbb{R}^d)} \eqsim \|(t,x) \mapsto \sqrt{t}\, \nabla \Gamma(t,0) f(x)\|_{T^{p,2}_\sigma}. \tag{1}$$

At this stage, we do not know if, more generally for $\sigma \geq 0$,

$$\|(-\Delta)^{\frac{\sigma}{2}} f\|_{L^p(\mathbb{R}^d)} \eqsim \|(t,x) \mapsto \sqrt{t} \nabla \Gamma(t,0) f(x)\|_{T^{p,2}_\sigma}.$$

As written for every $\sigma \geq 0$, the answer is no in the simplest possible case.

Theorem 1. *Take identity coefficients, so that $\Gamma(t,0) = e^{t\Delta}$, and take $p = 2$. For every $\sigma \geq 1$ and every nonzero $f \in \mathcal{S}(\mathbb{R}^d)$, the left side of (1) is finite and the right side is infinite. Consequently (1) fails for every $\sigma \geq 1$ even for the ordinary heat equation.*

2 The endpoint obstruction

At $p = 2$, Fubini removes the spatial averaging in the tent norm. Therefore

$$\|(t,x) \mapsto \sqrt{t}\, \nabla e^{t\Delta} f(x)\|_{T^{2,2}_\sigma}^2 = \int_0^\infty t^{-\sigma} \|\nabla e^{t\Delta} f\|_2^2 \, dt. \tag{2}$$

Proof of Theorem 1. The heat semigroup is strongly continuous on $H^1(\mathbb{R}^d)$, and hence

$$\nabla e^{t\Delta} f \longrightarrow \nabla f \quad \text{in } L^2(\mathbb{R}^d) \qquad (t \downarrow 0).$$

A nonzero L^2 Schwartz function cannot have zero gradient. Thus there are $c, \delta > 0$ such that $\|\nabla e^{t\Delta} f\|_2^2 \geq c$ for $0 < t < \delta$. Formula (2) is bounded below by

$$c \int_0^\delta t^{-\sigma} \, dt = \infty \qquad (\sigma \geq 1).$$

On the other hand, $f \in \mathcal{S}$ implies $(-\Delta)^{\sigma/2} f \in L^2$ for every $\sigma \geq 0$. $\square$

This is not a rough-coefficient pathology: identity coefficients satisfy all ellipticity and measurability hypotheses of the source. Nor is it an endpoint convention: for every $\sigma > 1$ the divergence is stronger.

3 The sharp positive range for the heat model

The same calculation identifies the exact threshold and verifies all powers.

Proposition 1. *For* $0 \le \sigma < 1$ *and* $f \in \mathcal{S}(\mathbb{R}^d)$,

$$\|(t,x) \mapsto \sqrt{t}\,\nabla e^{t\Delta} f(x)\|^2_{T^{2,2}_\sigma} = 2^{\sigma-1}\Gamma(1-\sigma)\|(-\Delta)^{\sigma/2} f\|_2^2. \tag{3}$$

Proof. By Plancherel, Tonelli, and $\lambda = |\xi|^2$,

$$\begin{aligned}\int_0^\infty t^{-\sigma}\|\nabla e^{t\Delta} f\|_2^2 dt &= \int_{\mathbb{R}^d} |\widehat{f}(\xi)|^2 |\xi|^2 \left(\int_0^\infty t^{-\sigma} e^{-2t|\xi|^2} dt\right) d\xi \\ &= 2^{\sigma-1}\Gamma(1-\sigma)\int_{\mathbb{R}^d} |\xi|^{2\sigma} |\widehat{f}(\xi)|^2 d\xi.\end{aligned}$$

The gamma integral is finite exactly when $\sigma < 1$. □

Thus a corrected first-order formulation can only ask about $0 \le \sigma < 1$. At and above one, a square-function kernel with additional small-time cancellation (for example, more powers of $t(-\Delta)$) is needed. This also shows that the all-σ "change of square function" heuristic immediately preceding the source question cannot apply to the displayed first-order square function at $\sigma \ge 1$.

4 Verification, novelty, and limitation

The included script checks the subcritical gamma identity and the logarithmic and power divergences of truncated time integrals. These computations are sanity checks; equations (2)–(3) are the proof.

A bounded search on 13 August 2026 used the exact formula, paper title, and tent/Sobolev terminology. It found both the arXiv and published source, which repeat the same question, but no later record of this correction. The novelty verdict is therefore *apparently new within the bounded search*, not an exhaustive bibliographic claim.

The theorem fully answers the question as quantified over all $\sigma \ge 0$. It does not decide the corrected and substantially deeper problem for $0 < \sigma < 1$, rough time-dependent coefficients, and general p.

References

[1] P. Portal and M. Veraar, *Stochastic maximal regularity for rough time-dependent problems*, Stoch. Partial Differ. Equ. Anal. Comput. **7** (2019), 541–597; arXiv:1810.01183, p. 35; doi:10.1007/s40072-019-00134-w.